

A Vision-Based Approach to Early Detection of Drowning Incidents in Swimming Pools

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Abstract—We present in this paper a vision-based approach to detection of drowning incidents in swimming pools at the earliest possible stage. The proposed approach consists of two main parts: a vision component which can reliably detect and track swimmers in spite of large scene variations of monitored pool areas, and an event-inference module which parses observation sequences of swimmer features for possible drowning behavioral signs. The vision component employs a model-based approach to represent and differentiate the background pool areas and foreground swimmers. The event-inference module is constructed based on a finite state machine, which integrates several reasoning rules formulated from universal motion characteristics of drowning swimmers. Possible drowning incidents are quickly detected using a sequential change detection algorithm. We have applied the proposed approach to a number of video clips of simulated drowning and obtained promising results as reported in this paper.

Index Terms—Background modeling, drowning incident, human motion analysis, segmentation and tracking, sequential change detection.

I. INTRODUCTION

THIS paper presents a novel vision-based approach to early detection of possible drowning incidents in swimming pools. There are numerous swimming pools worldwide, and drowning is one of the leading causes of accidental death. According to the Centers for Disease Control and Prevention, 4406 people drowned in the United States in 1998 [1]. Many drowning incidents occurred in public swimming pools staffed with professional lifeguards. Even more swimmers in near-drowning accidents are left with irremediable injuries, mostly to their brains, due to the lack of timely rescues. Therefore, there is a need for automated drowning-detection systems to provide useful assistance to lifeguards on duty by drawing their attention to possible drowning or to enhance the safety of unattended pools by alerting people nearby to accidental drowning.

Several drowning-detection systems, which use a variety of sensors to monitor swimmers, can be found in patents granted over the past twenty years [2]–[7]. The methods proposed in [2] and [3] use an array of ultrasonic transducers to detect a suspicious rate of swimmer motion, assumed to be the sign of possible drowning. The system in [4] proposes using ultrasonic radar to scan the pool bottom for motionless bodies. The ap-

proaches taken by [5]–[7] rely on special devices worn by swimmers to evaluate their conditions based on either the duration of a swimmer remaining below a certain depth underwater or some physiological signals such as heartbeat and breathing rates.

As the aforementioned systems involve cumbersome or intrusive sensor devices, systems relying on video cameras appear to be advantageous. One of the main advantages of using video cameras is that lifeguards have access to the video data associated with a possible drowning so that they can immediately inspect the swimmer's condition. This is particularly important to this problem, as the lifeguards are expected to make better judgment than the current state-of-the-art automated systems on whether a drowning incident has occurred.

Existing camera-based drowning-detection systems are surprisingly rare except for a few reported in patents [8], [9]. In general, these systems make use of underwater cameras to detect motionless bodies at the bottom of a swimming pool; drowning is assumed if a body is detected to be moving slowly or remaining still underwater over a prolonged duration. Although helpful in some cases, these systems leave much to be desired. First, none of these systems examines early drowning behavioral signs, such as struggles on water surface, which we consider important cues for both reliable detection and timely rescue of drowning swimmers. Second, these systems only rely on lack of motion to detect drowning incidents; false alarms could be triggered by swimmers merely standing or gently treading water. To reduce such false alarms, some systems are installed deep in water. However, by the time the body sinks down to such depth, the swimmer is likely at the late stage of drowning and little time is left for rescue. Third, the use of underwater cameras not only incurs high installation and maintenance cost but also faces the problem of cameras being easily occluded by nearby swimmers, objects, or dirt in the water.

The proposed approach is developed to address some of the limitations mentioned above. It consists of two main parts.

- 1) A vision component makes use of an overhead camera to detect and track swimmers in spite of nonstationary nature of swimming pools. Robust motion and shape features of the tracked swimmers are then extracted to form observation sequences for evaluating the swimmers' conditions.
- 2) An event-inference module makes use of a sequential change detection algorithm to parse the observation sequences of swimmer features for drowning behavioral signs. The module is constructed based on a finite state machine, which integrates several reasoning rules formulated from universal motion characteristics of drowning swimmers.

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As the proposed approach is designed to identify swimmers exhibiting specific motion characteristics, i.e., the drowning behavioral signs, it falls into the research area of human motion analysis. Typical human motion analysis consists of a low-level vision component for monitoring objects of interest and a high-level reasoning module for interpreting the outputs from the vision component. Interested readers are referred to [10] and [11] for more comprehensive reviews of research on human motion analysis.

Depending on the target environments, the difficulty of the task for the vision component can vary widely. As the proposed approach operates in a swimming pool where the scene changes are caused by not only swimmers but also other motion clutters such as water splashes and ripples, it requires a background model capable of coping with such a nonstationary environment. We briefly review here a few background modeling methods which have been proposed to handle different types of background scenes. The work of W4 [12] models the background scene by representing each background pixel with its minimum intensity, maximum intensity, and maximum intensity difference between consecutive frames. This method is devised to work mainly with stationary background scenes. Ahmed *et al.* [13] propose to suppress motion clutters by comparing a detected foreground pixel with the distributions of other background pixels within a small neighborhood. This attempt manages to suppress small motion clutters such as those caused by movements of tree branches and bushes. As a small circular neighborhood with a diameter of five pixels is used in their implementation, the scheme aims to handle relatively small motion. On the other hand, an increase in the neighborhood size will incur higher computational cost and reject valid foreground pixels. The work of [14] and [15] constructs statistical background models by approximating the color distribution of each background pixel with a mixture of Gaussian distributions. In [14], the background pixels are explicitly classified into three types corresponding to road, shadow, and vehicles, and a Gaussian mixture model is estimated for each pixel using an incremental expectation-maximization (EM) algorithm, while in [15] an online k -means approximation is used to estimate and update the Gaussian mixture model. These two approaches bear some similarity to our vision component. However, to deal with large shifts of background pixels due to water ripples and to take advantage of scene homogeneity of swimming pools, we model the entire monitored pool area, instead of each background pixel, with a mixture of Gaussian distributions. This approach also reduces the amount of memory required to store the background model and makes the updating of the background model easier.

Many high-level reasoning modules have also been designed to recognize specific types of human motions or gestures. For example, Yamato *et al.* [16] use a hidden Markov model (HMM) to differentiate several tennis actions, while Starner *et al.* [17] employ a set of HMMs to recognize hand signs of American Sign Language. As some human motions have rather deterministic sequential structures, Darrell *et al.* [18] and Bobick *et al.* [19] use dynamic time warping to match sequences of extracted features to prestored motion templates. In [20], the spatial and temporal characteristics of human motions are captured using

motion-energy image and motion-history image. Oliver *et al.* [21] use a coupled HMM as a behavior model to recognize five types of human interactions.

In our case, however, the human motions of interest are those that can provide prompt and sufficient evidence indicating the occurrence of a possible drowning. Although there has been a large amount of research effort [16]–[21] on human motion analysis, few efforts have put emphasis on studying human motions that respond to potentially dangerous situations, such as drowning. The main difficulty is due to the lack of examples of these anomalous motions exhibited by people in danger. To deal with the problem of limited training data, Oliver *et al.* [21] develop a framework for creating synthetic agents to mimic human motions and then use the synthetic data to train a behavioral model for recognizing different human interactions. The resulting system has been applied to a few test videos of real pedestrian scenes, and promising results have been obtained.

In [22], Picard *et al.* conduct experiments to develop techniques for recognizing human affective states, such as grief, anger, and joy, from four physiological signals. In their experiments, as each individual may have different interpretations of the same emotion, the physiological signals are gathered from a single subject with the aids of descriptive guidelines in order to obtain consistent physiological patterns. This work provides two useful insights: 1) in developing a system for inferring human emotion or behavior, it is possible to make use of simulation data collected from a limited number of subjects and 2) the methodology used for collecting and analyzing the data should not depend much on the subjects, so that it can be later generalized to develop a complete system. Using the similar principle, we have developed and tested the proposed approach using video data of swimmers, who imitated the behaviors of drowning swimmers according to a documentary video of drowning incidents and under the guidance of a professional lifeguard.

The remainder of the paper is organized as follows. The problem formulation of the proposed approach is given in Section II. Section III presents the low-level vision component for detecting and tracking swimmers, and Section IV describes the robust motion and shape features extracted from the swimmers. The event-inference module is detailed in Section V, and experimental results are reported in Section VI. In Section VII, we summarize our work and sketch future research directions.

II. PROBLEM FORMULATION

Our goal is to develop a vision-based approach that is capable of detecting possible drowning incidents at the earliest possible stage using only off-the-shelf overhead cameras. To achieve this goal, the following problems are to be addressed.

- How do we detect the presence of swimmers and track them over time in a swimming pool environment?
- What features can be extracted to characterize swimmers' behaviors?
- How do we infer a drowning incident? Or what rules can be used to determine if a swimmer has lapsed into a drowning situation?

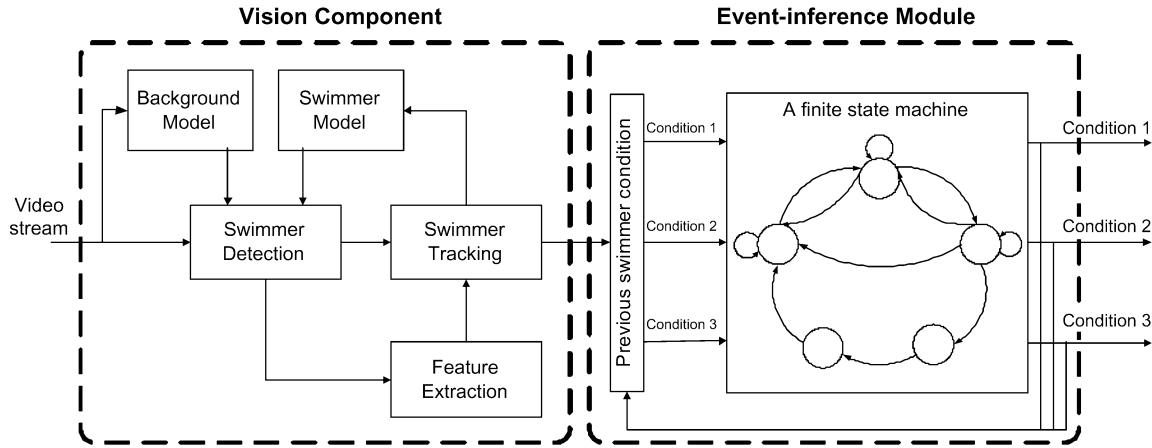


Fig. 1. Architecture of the proposed approach. There are three different swimmer conditions: 1) normal, 2) treading watering, and 3) drowning. Depending on the previous swimmer condition, a finite state machine evaluates the current swimmer condition based on various reasoning rules.

- How do we identify a drowning incident as soon as it occurs?

We have developed our approach by exploiting the learning from professional water crisis recognition. Beneath the contemporary water crisis recognition theory lies one fundamental question: what behavioral signs of a swimmer would alert lifeguards on duty that the swimmer has lapsed into a drowning situation? Prior work shows that a lifeguard's scanning for drowning swimmers is mainly behavior-centered, as "the evaluation of a swimmer's condition must always be based on the swimmer's behavior, not on other characteristics" [23]. Furthermore, to experienced lifeguards, the behavioral signs of a drowning swimmer, such as struggles on water surface, are "in a highly predictable, patterned, and recognizable way" [24]. These behavioral signs, referred to as Instinctive Drowning Responses (IDR) in [24], are universal responses of drowning swimmers to the actual or perceived feeling of suffocation in water.

Depending on whether they struggle before sinking to the bottom of swimming pools, drowning swimmers can be generally grouped into two types—active and passive drowning swimmers. Caused by a sudden loss of consciousness, a passive drowning swimmer normally slips under water without much struggle. Hence, a sudden lack of discernible movement can be used as the main criterion to detect passive drowning swimmers. There is, however, no single or simple criterion that can be used to reliably and timely identify an active drowning swimmer. In a study on the behavioral signs of drowning swimmers, Pia [24] summarizes the motion characteristics of active drowning swimmers as follows:

- 1) They are instinctively forced to laterally extend their arms and press down the water surface for support.
- 2) They do not have voluntary control of their limbs, but usually struggle to keep afloat on water surface for about 20 to 60 s.
- 3) Their bodies remain vertical in water, and they are not able to move in a horizontal or lateral direction.

These motion characteristics are commonly used by lifeguards today as the most important visual criteria to identify drowning

swimmers. We therefore decided to make use of these motion characteristics to evaluate the swimmers' conditions.

Fig. 1 shows the architecture of the proposed approach. It starts with building a statistical background model of the monitored pool area. This is achieved by using a mean-shift clustering algorithm [25] to identify distinct scene classes of the pool, and followed by an EM algorithm [26] to model the entire monitored pool area with a mixture of Gaussian distributions. The presence of swimmers can be detected by identifying sizable image regions which do not conform to this background model. The pixels of a newly detected swimmer are then accumulated to construct a swimmer appearance model, which will be used to segment the swimmer in subsequent frames. While being tracked, each swimmer has several robust shape and motion features to be extracted into observation sequences. An event-inference module is used to parse the observation sequences of swimmer features to determine whether a swimmer has lapsed into a drowning situation. The function blocks shown in Fig. 1 will be further elaborated in Sections III–VI.

III. VISION COMPONENT

The function of the vision component is to detect and track multiple swimmers in the monitored pool area, as well as to extract their motion and shape features into observation sequences. Due to swimmer movements, wind blow, and sunlight reflection, the nonstationary nature of a swimming pool precludes direct application of existing object/motion detection and segmentation methods, including those based on background subtraction [27], [28], temporal differencing [29], and optical flow computation [30]. The main reason is that, when swimmers are moving about in a pool, the intensity or color values of some background pixels can change quickly and substantially over successive frames; for instance, in the typical example shown in Fig. 2, a pixel can be the image of black stripes on the pool floor at one frame, water at another frame, and ripple at a subsequent frame. It is therefore rather difficult to separate swimmers from the nonstationary background scene of a swimming pool. To work with such a challenging environment, the vision component is developed by exploiting the scene homogeneity of a

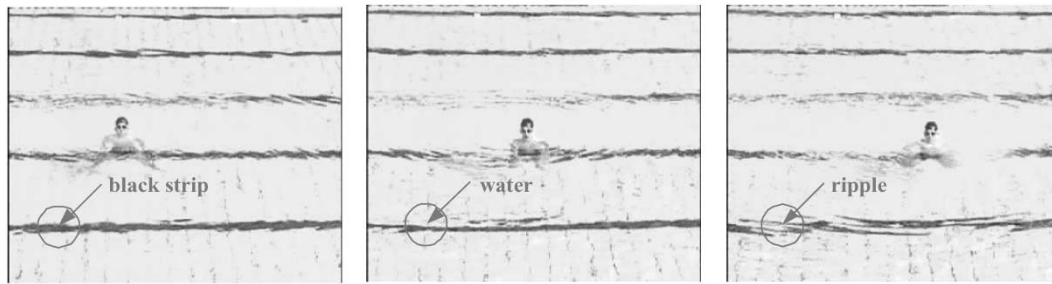


Fig. 2. Sample frames showing large scene variation caused by swimmer movements and water ripples in a swimming pool.

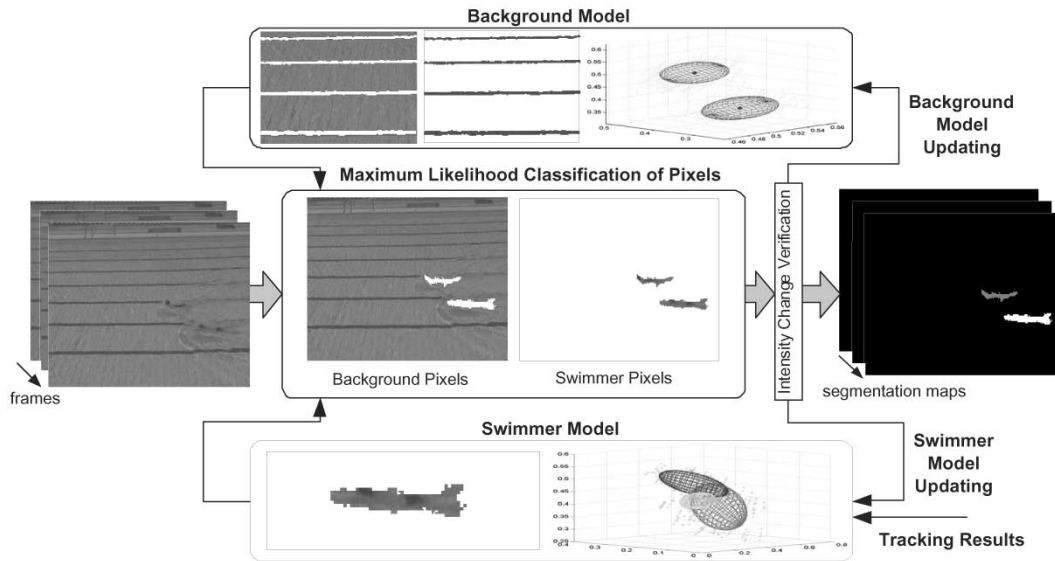


Fig. 3. Block diagram of the proposed swimmer detection and segmentation method.

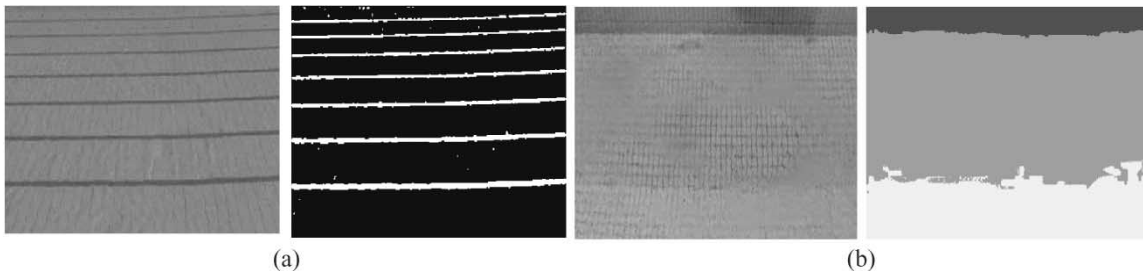


Fig. 4. Background scenes of two swimming pools and the dominant scene classes identified.

swimming pool and constructing swimmer appearance models as follows.

A. Swimmer Detection

In our approach, a high-mounted overhead camera with an oblique view is used to monitor swimmers in a pool. The swimmers are detected, segmented, and tracked from the video captured by this camera. Fig. 3 shows the block diagram of the proposed swimmer detection and segmentation method. The operation of each function block is further explained below.

Background Modeling: Unlike many other natural scenes, the background scenes of most swimming pools are rather

homogeneous, consisting of only a few distinct scene classes. Here, a scene class refers to a group of pixels that have similar color; typical examples include pixels of the water or those of the black stripes on the floor [see Fig. 4(a)].

In the proposed method, the background scene of the entire monitored pool area is modeled with a multivariate Gaussian mixture model (GMM), of which each Gaussian component accounts for the color distribution of one scene class. To illustrate the reason for using GMM, the hue, saturation, and value (HSV) color values of pixels belonging to two different scene classes, the water and the black floor stripes of a swimming pool, are shown in the “Background Model” function block in Fig. 3. It can be seen that pixels of each scene class tend to form a tight

cluster in the HSV color space and fit well to a multivariate Gaussian distribution.

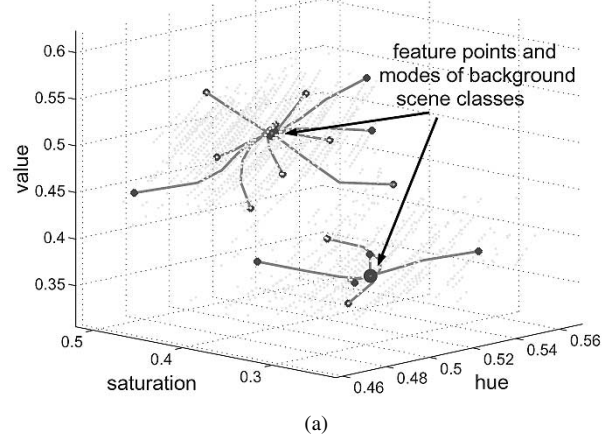
The standard EM algorithm [26] is used to estimate the parameters of the Gaussian mixture model. To provide the EM algorithm with the number of Gaussian components and the initial estimate of the model parameters, a mean-shift clustering algorithm [25] is used to identify the dominant scene classes of the monitored pool area. Let the HSV color values (in the range 0–1) of the swimming pool's pixels be the feature points to be grouped into different scene classes. The mean-shift clustering searches for the local modes of feature points by calculating the mean-shift vector of each feature point X (the HSV color value of a pixel) from a neighborhood region covered by a uniform kernel $S_h(X)$ of width h [25]. Specifically, the mean-shift vector at X is given by

$$M_h(X) = \frac{1}{n_X} \sum_{X_i \in S_h(X)} X_i - X \quad (1)$$

where n_X denotes the number of feature points inside the kernel $S_h(X)$. It is shown in [25] that, when the density of feature points within the kernel is not uniform, the mean-shift vector always points toward the direction of the maximum increase in density, thus defining a path leading to a local maximum (or mode) of the density. A mode-seeking procedure, originating from dense samples of feature points, can thus be devised; it consists of 1) calculating the mean shift vector $M_h(X)$; 2) translating the kernel $S_h(X)$ at X by $M_h(X)$, i.e., $X \leftarrow X + M_h(X)$; and 3) repeating the procedure for subsequent X until the magnitude of $M_h(X)$ is sufficiently close to zero. Fig. 5(a) illustrates an example of finding the modes of the dominant scene classes in the background scene shown in Fig. 4(a); the solid (red) lines show the trajectories of the kernel translations starting from a number of initial sample points indicated with dark (blue) dots. It can be seen that feature points belonging to the same dominant scene class gravitate toward the same convergence point: the mode of the scene class.

Once the modes of all the dominant scene classes have been located, the background pixels are grouped into different scene classes by assigning each pixel to the class whose mode has the most similar color to that of the pixel. The number of resultant scene classes depends on the kernel width h (which is set to 0.05 in our implementation); however, the robustness of the mean-shift clustering and the scene homogeneity of swimming pools allow us to reliably identify the dominant scene classes of different pools without changing the kernel width. Fig. 4 shows two typical background scenes of swimming pools and their dominant scene classes (shown in different intensities) found by the mean-shift clustering. After obtaining the dominant scene classes, the standard EM algorithm is used to estimate the parameters of the GMM for the background scene. The number of Gaussian components and the initial estimate of the model parameters are derived from the resultant dominant scene classes. Specifically, the number of Gaussian components is set to the number of scene classes, while the initial model parameters (i.e., the mean vector, covariance matrix, and mixing proportion) of each Gaussian component are computed from pixels in the corresponding scene class. A good initial estimates can usually

Finding dominant scene classes using mean-shift clustering



Background Model

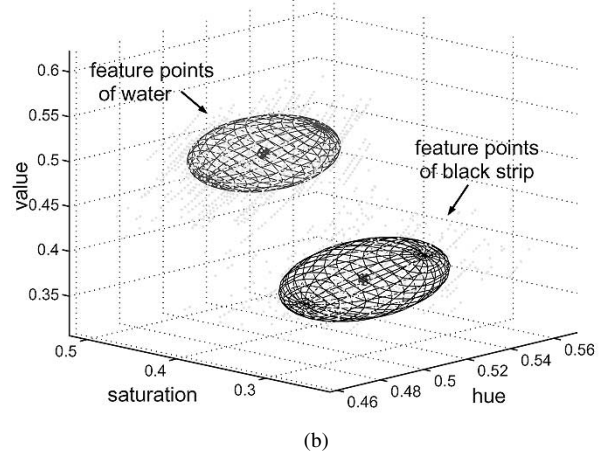


Fig. 5. (a) The dominant scene classes of the background scene shown in Fig. 4(a) identified by mean-shift clustering. (b) The corresponding GMM obtained by the EM algorithm.

be obtained by the mean-shift clustering for this type of background scenes; hence, the EM algorithm quickly converges to a model that can well represent the color distribution of the entire monitored pool area. Fig. 5(b) shows that there is a reasonable fit between each Gaussian component (where the ellipsoid shows the surface of equal probability density of the Gaussian component) and the feature points of the corresponding scene class.

Let $\mathcal{B}$ denote the background model consisting of a mixture of k Gaussian components $\{\mathcal{B}_i\}_{i=1}^k$, each characterized by its mean vector $\mu_{\mathcal{B}_i}$ and covariance matrix $\Sigma_{\mathcal{B}_i}$, in some proportions $\alpha_1, \dots, \alpha_k$, where $\sum_{i=1}^k \alpha_i = 1$ and $\alpha_i > 0$. The probability that a pixel j with HSV color value X_j belongs to the background model $\mathcal{B}$ can be computed as

$$\begin{aligned} p(X_j|\mathcal{B}) &= \sum_{i=1}^k \alpha_i p(X_j|\mathcal{B}_i) \\ &= \sum_{i=1}^k \frac{\alpha_i}{(2\pi)^{3/2} |\Sigma_{\mathcal{B}_i}|^{1/2}} \\ &\quad \times \exp \left\{ -\frac{1}{2} (X_j - \mu_{\mathcal{B}_i})^T \Sigma_{\mathcal{B}_i}^{-1} (X_j - \mu_{\mathcal{B}_i}) \right\}. \end{aligned} \quad (2)$$

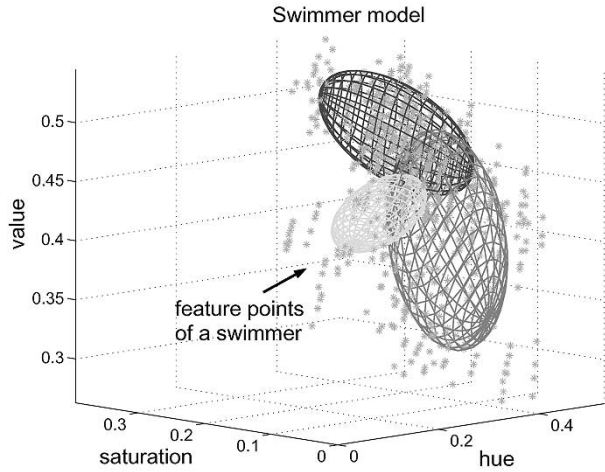


Fig. 6. GMM of a swimmer obtained by the EM algorithm.

If the probability $p(X_j|\mathcal{B})$ is less than a preset threshold, the pixel is considered a nonbackground pixel. All such pixels sharing some spatial proximity are then grouped together by a standard connected component algorithm [31]. The presence of a new swimmer is assumed if a sizable connected region of nonbackground pixels (100 pixels in our implementation) has been detected.

Swimmer Modeling: Once a sufficient number of pixels corresponding to a swimmer have been accumulated, a color appearance model for the swimmer is constructed. Similar to the background modeling, a Gaussian mixture model is used to model the color distribution of the swimmer pixels. That is, the color value of each swimmer pixel is viewed as a realization from a GMM $\mathcal{S}$ consisting of k Gaussian components $\{\mathcal{S}_i\}_{i=1}^k$, each characterized by its mean vector $\mu_{\mathcal{S}_i}$ and covariance matrix $\Sigma_{\mathcal{S}_i}$, in some proportions $\gamma_1, \dots, \gamma_k$, where $\sum_{i=1}^k \gamma_i = 1$ and $\gamma_i > 0$; the number of Gaussian components k is usually found to be 3 for most swimmers. Fig. 6 shows an example of swimmer model obtained from a test video. The probability that a pixel j with HSV color value X_j belongs to the swimmer model $\mathcal{S}$ can be computed as

$$\begin{aligned} p(X_j|\mathcal{S}) &= \sum_{i=1}^k \gamma_i p(X_j|\mathcal{S}_i) \\ &= \sum_{i=1}^k \frac{\gamma_i}{(2\pi)^{3/2} |\Sigma_{\mathcal{S}_i}|^{1/2}} \\ &\quad \times \exp \left\{ -\frac{1}{2} (X_j - \mu_{\mathcal{S}_i})^T \Sigma_{\mathcal{S}_i}^{-1} (X_j - \mu_{\mathcal{S}_i}) \right\}. \end{aligned} \quad (3)$$

Swimmer Segmentation: The swimmer segmentation is closely coupled with the swimmer tracking process, which will be presented in Section III-B. Since the position and the size of a tracked swimmer can only change smoothly over subsequent frames when the processing frame rate is sufficiently high, based on the segmentation and tracking results in the previous frame the vision component can define in the current frame a local search window where each particular swimmer is likely to appear. The segmentation of a tracked swimmer in the current

frame can then be done by maximum-likelihood classification of pixels within the local search window. Specifically, given the background model $\mathcal{B}$ and the swimmer model $\mathcal{S}$, a pixel with color value X is considered to be a swimmer pixel if

$$p(X|\mathcal{S}) > p(X|\mathcal{B}) \quad (4)$$

where $p(X|\mathcal{B})$ and $p(X|\mathcal{S})$ are computed from (2) and (3), respectively.

However, the pixels satisfying (4) may contain motion clutters, mainly resulting from water reflection and water splashes, which are not explicitly accounted for by either the background model or the swimmer model. To see this, Fig. 7(a) shows one sample frame of a swimmer splashing water, and Fig. 7(b) and (c) depicts the top and side views of the frame's intensity mesh plot, where the different gray levels indicate high- and low-intensity values, respectively. Clearly, there is a distinctive difference between the intensities of the swimmer and the water splashes; the swimmer causes significant decrease in intensity value, while the water splashes make the scene brighter. This can also be seen from the intensity histogram of the sample frame [Fig. 7(d)], where histogram bins of the swimmer and the water splashes reside at different sides of the average water intensity. Hence, a pixel j satisfying (4) is considered a valid swimmer pixel only if

$$\frac{I_W - I_j}{I_W} > \tau \quad (5)$$

where I_j is the intensity of pixel j and I_W is the average water intensity. As the pool mainly contains the water, the average water intensity I_W can be obtained from the mean intensity of the scene class having the largest sample size. In our implementation τ is set to 0.1, as a swimmer pixel is normally at least 10% darker than the average water intensity. The effect of the threshold τ on intensity change is shown in Fig. 7(c) and (d). Note that a similar scheme using intensity contrast has also been adopted in [31] to perform people detection and tracking in London underground stations.

In order to cater for changes over time, the parameters of the background model and each swimmer model are updated regularly (every 5 s in our experiments) using a method similar to that proposed in [14]. The underlying principle is that each Gaussian distribution is updated with only those pixels having high likelihood of being realized by the associated distribution.

B. Tracking of Swimmers

The vision component uses a first-order Kalman filter [32] to track each detected swimmer over frames. During the tracking, the position of the swimmer in every subsequent frame can first be predicted by the filter so that a local search window can be specified for swimmer segmentation. Furthermore, a best-fit ellipse is obtained to approximate the size of each tracked swimmer, as will be further explained in Section IV. The size of the local search window is defined by the length of the ellipse's major axis. To ensure that the local search window fully encloses the tracked swimmer, halves of the horizontal and vertical displacements predicted by the Kalman filter are included as the horizontal and vertical tolerances, respectively.

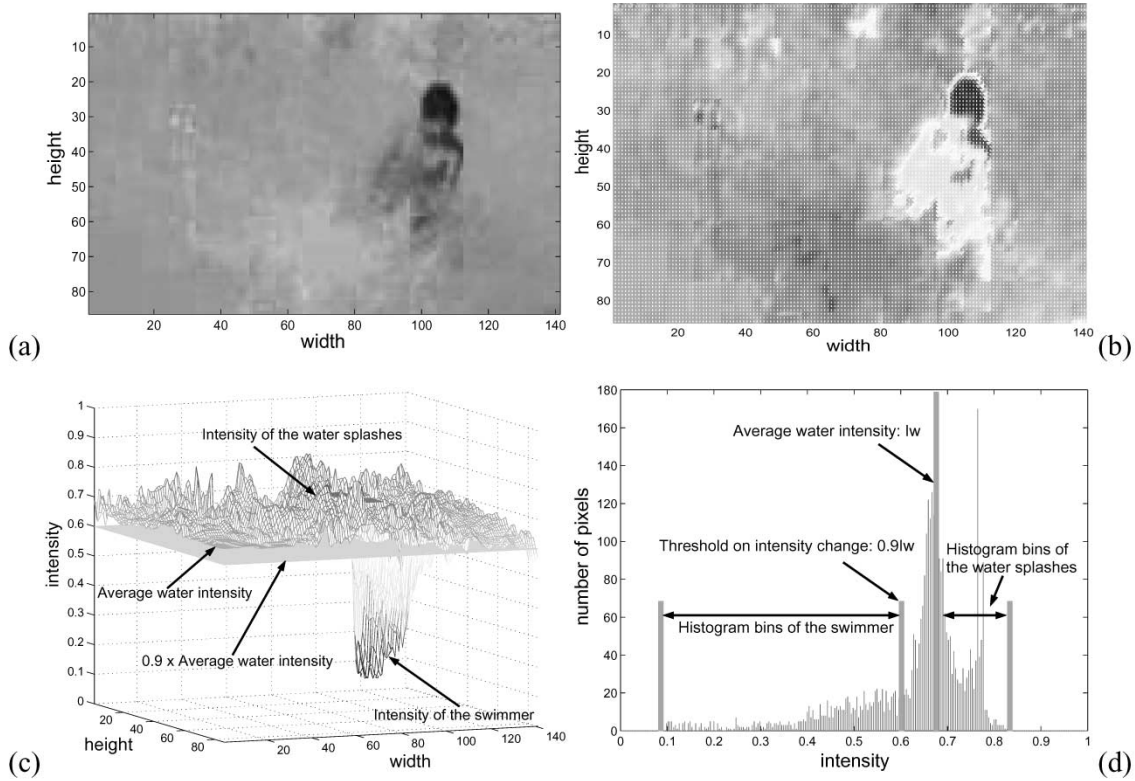


Fig. 7. (a) Sample frame of a swimmer splashing water. (b) Top view. (c) Side view of the intensity mesh plot. (d) The intensity histogram of the sample frame showing the difference between the intensities of the swimmer and the water splashes.

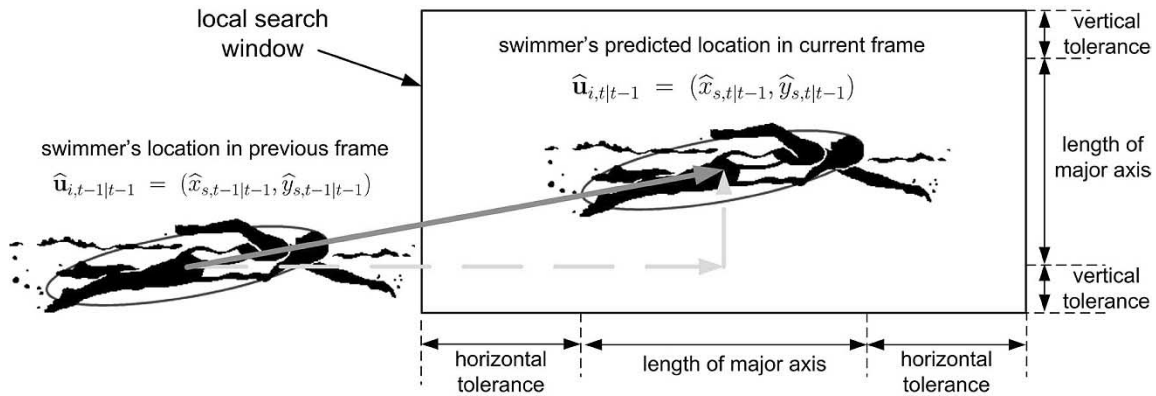


Fig. 8. Placement of a local search window for the segmentation of a tracked swimmer.

Fig. 8 illustrates how the local search window is defined for a tracked swimmer moving horizontally.

For each tracked swimmer, the state vector of the associated Kalman filter at an instance t is defined as $\mathbf{x}_t = (x_{s,t}, y_{s,t}, \Delta x_{s,t}, \Delta y_{s,t})^T$, where $(x_{s,t}, y_{s,t})$ are the x and y coordinates of the swimmer's location and $(\Delta x_{s,t}, \Delta y_{s,t})$ are the estimated horizontal and vertical interframe displacements of the swimmer's location. The Kalman filter uses the current measurement of the swimmer's location $\mathbf{z}_t$ to improve its prior estimate $\hat{\mathbf{x}}_{t|t-1}$ through a linear combination $\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t(\mathbf{z}_t - \mathbf{H}_t\hat{\mathbf{x}}_{t|t-1})$, where $\hat{\mathbf{x}}_{t|t}$ is the updated estimate, $\mathbf{H}_t$ is the observation matrix, and $\mathbf{K}_t$ is the Kalman gain at instance t . The prior estimate of

the state vector for the next instance $t + 1$ can be computed as $\hat{\mathbf{x}}_{t+1|t} = \mathbf{F}_t\hat{\mathbf{x}}_{t|t}$, where $\mathbf{F}_t$ is the system matrix. (The parameters of a first-order Kalman filter such as $\mathbf{H}_t$, $\mathbf{K}_t$, and $\mathbf{F}_t$ can be found in [32].)

The correspondence between a tracked swimmer in the previous frame and a segmented swimmer in the current frame is established by examining their locations, defined as the positions of their ellipses' centroids. For a tracked swimmer i , let his/her filtered location in the previous frame and the predicted location in the current frame be denoted as $\hat{\mathbf{u}}_{i,t-1|t-1} = (\hat{x}_{s,t-1|t-1}, \hat{y}_{s,t-1|t-1})$ and $\hat{\mathbf{u}}_{i,t|t-1} = (\hat{x}_{s,t|t-1}, \hat{y}_{s,t|t-1})$, respectively (see Fig. 8). The matching distance between the tracked swimmer i in the pre-

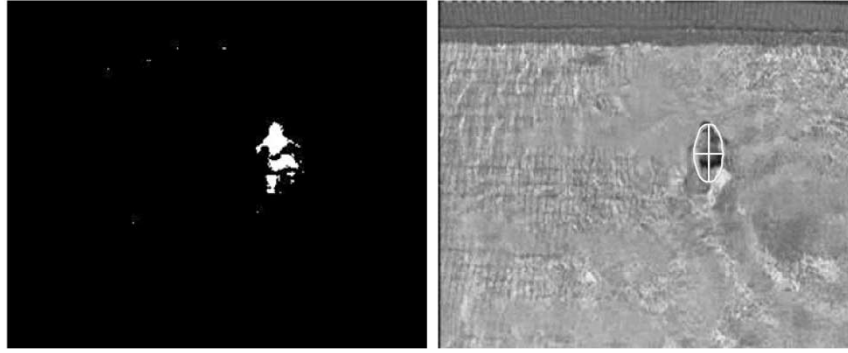


Fig. 9. Best-fit ellipse modeling fragmented segmentation result.

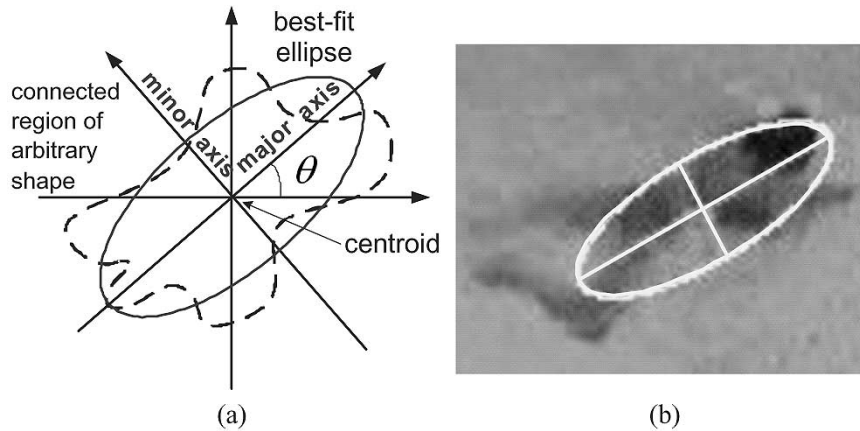


Fig. 10. Best-fit ellipse representing: (a) a connected region and (b) a segmented swimmer.

vious frame and a segmented swimmer j in the current frame, whose location is denoted as $\mathbf{u}_{j,t} = (x_{s,t}, y_{s,t})$, is given by

$$d_{ij} = \|\hat{\mathbf{u}}_{i,t-1|t-1} - \mathbf{u}_{j,t}\| + \|\hat{\mathbf{u}}_{i,t|t-1} - \mathbf{u}_{j,t}\|. \quad (6)$$

The segmented swimmer j is determined to be the swimmer i if $d_{ij} = \min\{d_{sj} | s = 1, 2, \dots, N\}$, where N is the total number of swimmers detected in the previous frame at instance $t - 1$, and d_{ij} is less than the maximum possible distance traveled by a swimmer within the interframe interval. Subsequently, the location of the matched swimmer j , i.e., $\mathbf{u}_{j,t}$, is taken as the measurement $\mathbf{z}_t$ for the swimmer i and used to obtain the updated estimate $\hat{\mathbf{x}}_{t|t}$.

Unmatched swimmers from the previous frame are further examined to determine if they are making their way out of the monitored pool area (i.e., their centroids lie close to the borders of the monitored pool area). If not, it is assumed that the swimmer has been left out during the segmentation. This can sometimes happen if segmented swimmers are so fragmented that they are removed as segmentation errors. In this case, the tracking process is continued based on the prediction made by the Kalman filter. Unmatched segmented swimmers in the current frame are hypothesized as newly appearing swimmers and will be added to the tracking list after being tracked over several frames. To avoid possible mismatching due to swimmer collision, the vision component stops tracking the swimmers involved in a collision and resumes tracking only when they separate. Each swimmer is then treated as a newly appearing swimmer.

IV. FEATURE EXTRACTION

For each swimmer being tracked, his/her shape and motion features are extracted from the corresponding segmentation result. The sequences of these features will be parsed by an event-inference module (described in Section V) to evaluate the swimmer's condition.

The features are selected based on the output available from the vision component and the requirements of the event-inference module. Not only should the features be robust or tolerant to imperfect segmentation results such as over or under segmentation due to the nonstationary nature of swimming pools, they must also capture sufficient drowning behavioral signs so that the event-inference module can perform its task. With these considerations in mind, the following swimmer features, including shape feature (elongation measure), global motion feature (moving speed), and local motion feature (size variation), are extracted from every M consecutive swimmer observations. Here, a swimmer observation refers to the segmented swimmer in each frame, and M is set to 8 which is equal to the number of frames processed every second.

Shape Feature: Elongation Measure: The shape feature of a swimmer is extracted by fitting the segmented swimmer in each frame with a best-fit ellipse [31]. Other shape descriptors, such as rectangular bounding box, convex hull, and polygon, have also been examined, but the ellipse is found to be best suited to handle fragmented segmentation results (one example is shown in Fig. 9) and represent the swimmer's body posture (see Fig. 10).

In general, swimmers have two common body postures, being nearly vertical in water or parallel to water surface. As observed by an overhead camera, these two body postures can be characterized by the elongation measure of the best-fit ellipse around each swimmer. The elongation measure, defined as the ratio between the lengths of the major and minor axes of the ellipse, can be computed as [33]

$$\mathcal{EL} = \sqrt{\frac{I_{\max}}{I_{\min}}} \quad (7)$$

where $I_{\max}$ and $I_{\min}$ are the greatest and least moments of inertia for the ellipse. These two moments are calculated as

$$I_{\min} = \sum_{(x,y) \in \mathcal{R}} [(y - y_c) \cos \theta - (x - x_c) \sin \theta]^2 \quad (8)$$

$$I_{\max} = \sum_{(x,y) \in \mathcal{R}} [(y - y_c) \sin \theta + (x - x_c) \cos \theta]^2 \quad (9)$$

where $\mathcal{R}$ is the set of x and y coordinates of all the pixels belonging to the segmented swimmer, (x_c, y_c) is the ellipse's centroid defined as

$$x_c = \frac{1}{N} \sum_{(x,y) \in \mathcal{R}} x, \quad y_c = \frac{1}{N} \sum_{(x,y) \in \mathcal{R}} y \quad (10)$$

where N is the total number of pixels in set $\mathcal{R}$ and $\theta = 1/2 \tan^{-1}(2\mu_{1,1}/\mu_{2,0} - \mu_{0,2})$ is the angle of the major axis, where $\mu_{p,q}$ denotes the (p, q) -order central moments, given by

$$\mu_{p,q} = \sum_{(x,y) \in \mathcal{R}} (x - x_c)^p (y - y_c)^q. \quad (11)$$

To account for imperfect segmentation, the elongation measure is averaged over M consecutive swimmer observations. When the elongation measure is close to 1, it indicates that the segmented swimmer has a circular shape and the swimmer likely has a vertical body posture. On the other hand, when the elongation measure is much larger than 1, it shows that the swimmer maintains his/her body nearly parallel to the water surface.

Global Motion Feature: Moving Speed: The moving speed of a tracked swimmer is approximated by the displacement of his/her ellipse's centroid $\mathbf{u}_c = (x_c, y_c)$ over successive frames. To obtain a reliable measure of moving speed, especially when the swimmer is moving slowly, the displacement of the ellipse's centroid is computed as

$$\mathcal{MS} = \frac{2}{M} \sum_{i=1}^{M/2} \|\mathbf{u}_{c,i+M/2} - \mathbf{u}_{c,i}\| \quad (12)$$

from every M consecutive swimmer observations.

Local Motion Feature: Size Variation: The struggle of a drowning swimmer can be characterized by his/her fast limb movements. The movements cause the size of the segmented swimmer to change rapidly within a short duration. Hence, a normalized standard deviation of sizes for every M consecutive

swimmer observations can be used to characterize the local motion of the swimmer, defined as

$$\mathcal{SV} = \frac{\sqrt{\frac{1}{M} \sum_{i=1}^M (s_i - m_s)^2}}{m_s}$$

and

$$m_s = \frac{1}{M} \sum_{i=1}^M s_i \quad (13)$$

where $\{s_i\}_{i=1}^M$ is the set of sizes of M swimmer observations, each of which can be computed from the zeroth moment $\mu_{0,0}$ of the corresponding segmented swimmer.

V. EVENT-INFERENCING MODULE

In this section, an event-inference module is developed to quickly detect, from the observation sequences of swimmer features extracted by the vision component, unusual patterns indicating that the swimmer has lapsed into a drowning situation.

A. Rules for Evaluating Swimmer Condition

We formulate the universal motion characteristics of drowning swimmers, which have been described in Section II, as the following rules for detecting an active drowning swimmer, who:

- **Rule I:** Moves slowly or remains within a small neighborhood;
- **Rule II:** Has nearly vertical body posture in water;
- **Rule III:** Exhibits irregular and possibly fast limb movements.

Each of these rules is illustrated below using a video clip of simulated drowning. In the clip, a swimmer swims in breast stroke continuously till about frame 140 when he starts getting into difficulties.

Rule I: Test for Slow Moving Speed: This rule tests whether a tracked swimmer has a notable global motion. As discussed in Section IV, the interframe displacement of a swimmer's centroid is considered as his/her moving speed. Fig. 11(b) shows a video frame superimposed with consecutive swimmer observations, which are segmented and enclosed with black ellipses by the vision component. Fig. 11(a) depicts the sequences of the x and y positions of the swimmer's centroid in the video frames. It can be seen that the swimmer is confined within a small pool area after lapsing into a drowning situation.

Rule II: Test for Vertical Body Posture: An active drowning swimmer usually struggles to keep his/her mouth above water for breathing. As a result, the drowning swimmer has nearly vertical body posture in water, and only his/her head and probably a small part of the body above water are visible to the camera. This vertical body posture can be characterized by the nearly circular shape of the best-fit ellipse around the swimmer. On the other hand, if the swimmer is able to maintain the body almost parallel to the water surface, which is characterized by the elongated shape of the best-fit ellipse, the swimmer is less likely to be in danger. As can be seen from Fig. 12, the elongation measure of the best-fit ellipse drops significantly to a value close to 1 when the swimmer starts to drown.

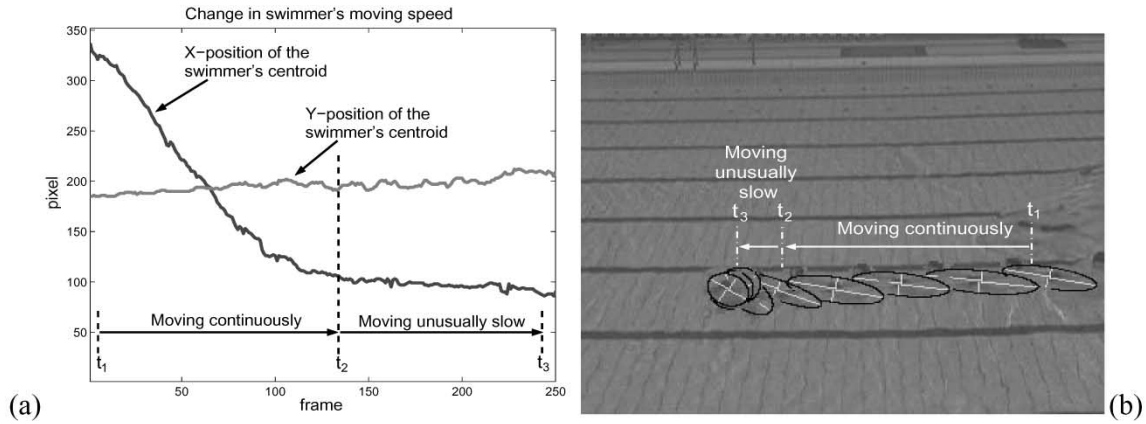


Fig. 11. Change in the swimmer's moving speed after lapsing into a drowning situation.

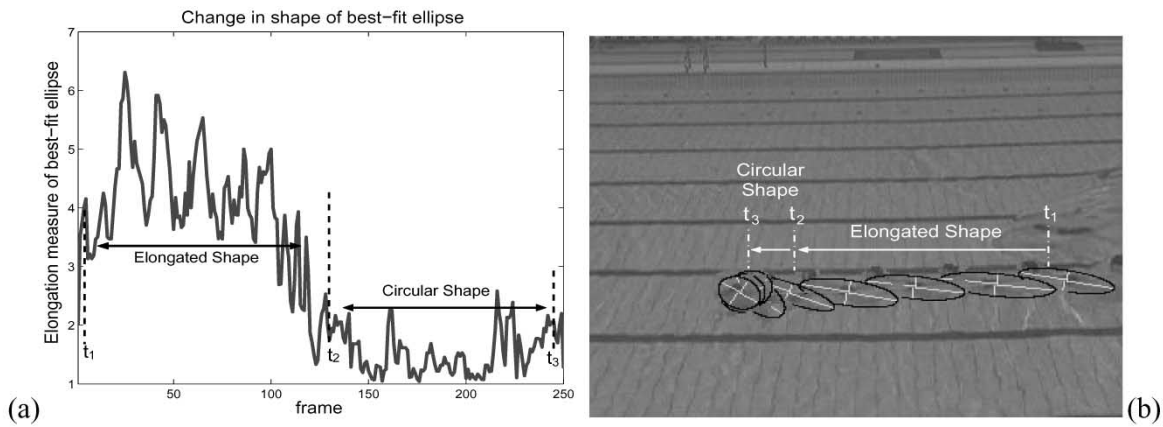


Fig. 12. Change in the swimmer's body posture after lapsing into a drowning situation.

Rule III: Test for Fast Limb Movements: The struggles of an active drowning swimmer, which can be characterized by his/her irregular and fast limb movements, normally result in a large variation of the size of the segmented swimmer within a short duration. Fig. 13(a) shows eight consecutive observations, indicated in light color, of a swimmer struggling on the water surface; the size of the segmented swimmer varies from the smallest, 455 pixels, to the largest, 805 pixels. In contrast, Fig. 13(b) shows eight consecutive observations of a swimmer treading water in another test video clip; the size of the segmented swimmer does not change significantly. The right plot of Fig. 13(c) depicts the time sequences of size variations of the segmented swimmers in these two test videos, which are extracted from the corresponding sequences of segmentation sizes shown in the left plot of Fig. 13(c). It can be seen that the drowning swimmer has much larger size variation than the swimmer treading water.

B. Integrating the Rules With a Finite State Machine

Table I lists three possible swimmer conditions considered in our approach, namely, normal, treading water, and drowning, which are categorized according to the characteristics of swimmers' motions and body postures. However, these three possible swimmer conditions may share some similar characteristics. For example, both a drowning swimmer and a swimmer treading

water exhibit slow moving speed and nearly vertical body posture; also, an energetic swimmer can have very fast limb movements while swimming freely in the pool.

To differentiate three swimmer conditions, we use a finite state machine to systematically integrate the rules formulated in Section V-A so that they can be properly applied to swimmers in different conditions (or states). The details are given as follows.

With reference to the state diagram shown in Fig. 14, the initial state of a newly appearing swimmer is assumed to be "Normal." In this state, Rules I and II are used to differentiate normal swimmers from drowning swimmers or swimmers treading water. Only when a swimmer is detected to be moving unusually slow and having nearly vertical body posture, i.e., Rules I and II are satisfied, the event-inference module suspects an onset of a possible drowning and changes the state of the swimmer to "Possible drowning/treading." In this new state, in addition to Rules I and II, the swimmer's condition will be further evaluated using Rule III. Only when all three rules are satisfied is the swimmer considered to be in the state of "Drowning." Alarms will be triggered after the swimmer remains in the "Drowning" state for a period longer than a preset duration.

In each state, the evaluation of a swimmer's condition can be formulated as a sequential hypothesis testing on a suitable sequence of swimmer features. Let $O = \{o_1 o_2 \dots o_T\}$ be the sequence of swimmer features obtained as described in Section IV;

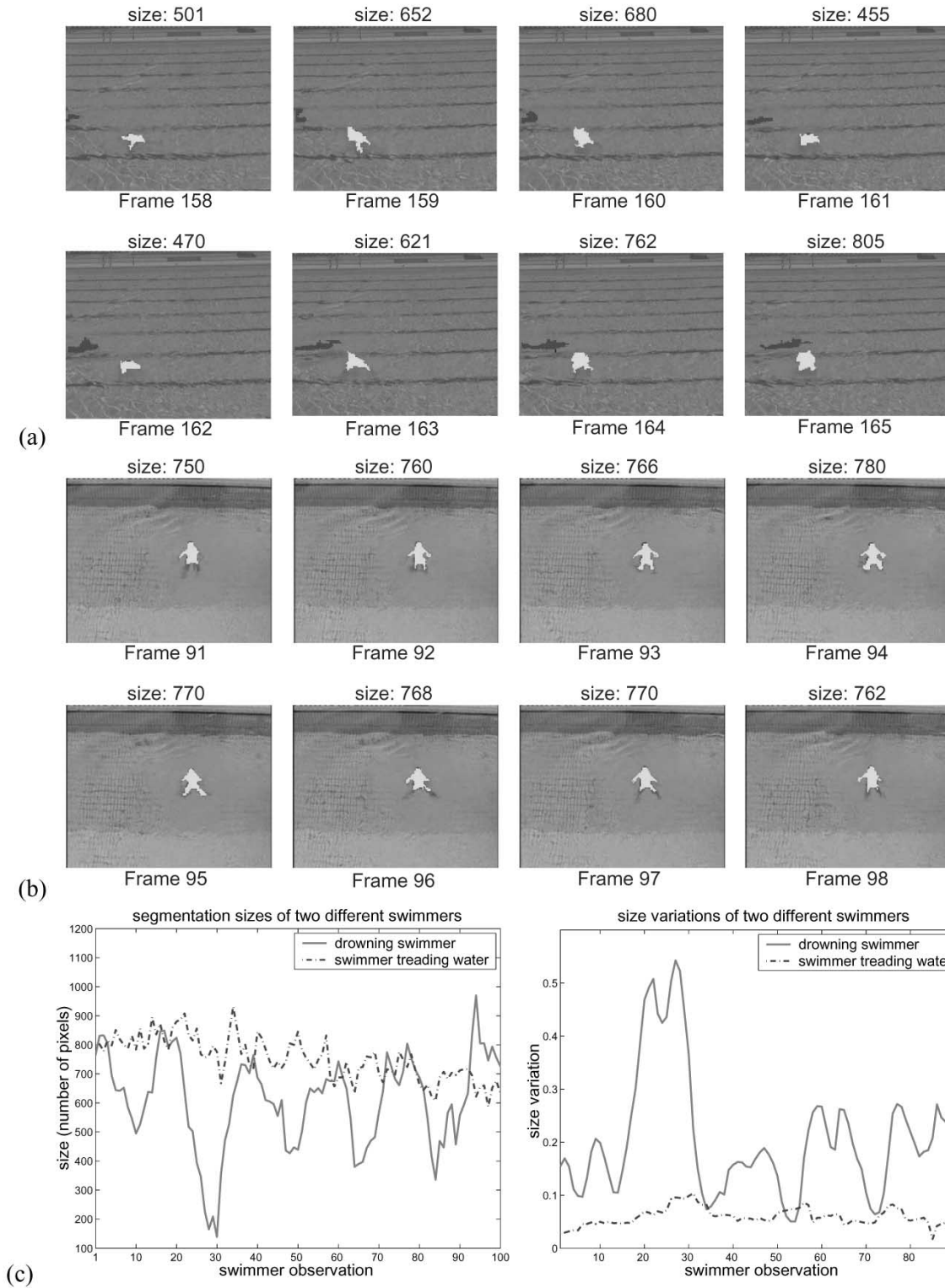


Fig. 13. Sample frames illustrating the size variation of: (a) a swimmer lapsing into a drowning situation, (b) a swimmer treading water, and (c) comparison of size variations between the two swimmers.

at any instance t , the feature vector o_t is assumed to be generated according to two possible underlying feature distributions, P_{w_1} and P_{w_2} , as illustrated in Fig. 15. The condition of a swimmer at each state can be determined by judging from which underlying distribution the associated swimmer features have likely been generated.

In our decision formulation as shown in Fig. 15, the detection of a drowning incident involves two types of state transitions, of which the associated swimmer features and their distributions are designated as $\{o_t|i, P_{w_1|i}, P_{w_2|i}\}_{i=1,2}$. For the type 1 state transition (based on Rules I and II), the swimmer's moving speed and elongation measure are of concern; hence $P_{w_1|1}$ and

TABLE I
THREE SWIMMER CONDITIONS CONSIDERED IN OUR APPROACH

Swimmer Features	Normal Swimmers		Swimmers Treading Water	Drowning Swimmers
	Type 1	Type 2		
Moving Speed	High	Low	Low	Low
Body Posture	Vertical/Parallel	Parallel	Vertical	Vertical
Limb Movement	Slow/Fast		Slow	Fast

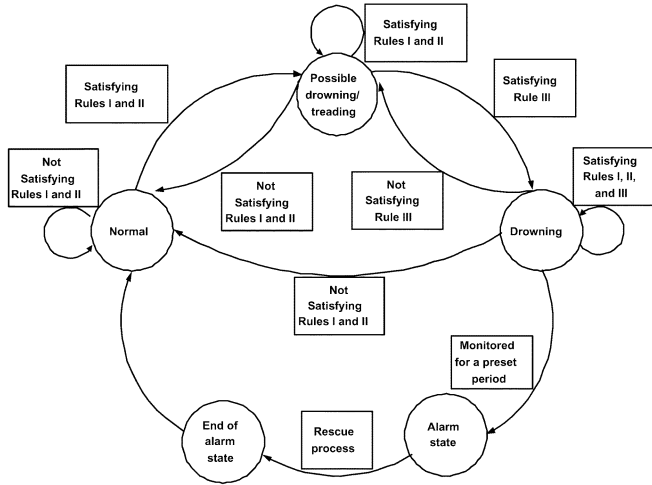


Fig. 14. Finite state machine integrating the formulated rules.

$P_{w_2|1}$ correspond to the distributions of $o_{t|1} = (\mathcal{MS}, \mathcal{EL})^T$ for normal swimmers ($w_1|1$) and drowning swimmers or swimmers treading water ($w_2|1$), respectively. For the type 2 state transition (based on Rule III), the swimmer's size variation is relevant, and $P_{w_1|2}$ and $P_{w_2|2}$ refer to distributions of $o_{t|2} = \mathcal{SV}$ for swimmers treading water ($w_1|2$) and drowning swimmers ($w_2|2$), respectively.

C. Swimmer Feature Modeling

To model the distributions of the relevant swimmer features at different states, denoted as $\{P_{w_1|i}, P_{w_2|i}\}_{i=1,2}$, from a training set, we first group all the training clips into three categories corresponding to the three swimmer conditions. For the clips in the drowning category, the vision component is used to extract swimmer features from frames containing drowning incidents identified manually by an experienced lifeguard. For the other two categories, the swimmer features are extracted from frames in which tracked swimmers are not close to the borders of the monitored pool area.

Subsequently, the feature distributions $\{P_{w_1|i}, P_{w_2|i}\}_{i=1,2}$ are modeled with GMMs using the EM algorithm. The number of Gaussian components in each mixture is determined based on the Bayesian information criterion [34], which strikes a good balance between the model's complexity and its accuracy in representing the feature distributions. Fig. 16 shows the scatter plot of swimmer features $o_{t|1} = (\mathcal{MS}, \mathcal{EL})^T$ and the histogram of another feature $o_{t|2} = \mathcal{SV}$ as well as their associated GMMs obtained from the training set. It can be seen that a certain degree of difference exists between the feature distributions under different swimmer conditions. Although at some instances the size variation of a drowning swimmer can be

as small as that of a swimmer treading water, and occasionally a normal swimmer may slow down to tread water for a short while, this ambiguity can be resolved by using the following sequential change-detection algorithm to accumulate sufficient evidence before declaring a change of swimmer condition.

D. Sequential Detection of State Transition

A sequential change detection algorithm is used to detect the change in the underlying feature distributions and thus the change of swimmer condition. For the sake of simple notation, the detection algorithm given below denotes the feature distributions of two swimmer conditions at a particular state as P_{w_1} and P_{w_2} , respectively. The detection of an abrupt change from one distribution P_{w_1} to the other distribution P_{w_2} can be stated as: given an observation sequence of features $O = \{o_1 o_2 \dots o_T\}$, there is an unknown abrupt change time t_c such that the features are generated according to the distribution P_{w_1} before t_c and the distribution P_{w_2} after t_c . The objective is to detect this abrupt change time t_c , if it exists, as soon as possible while minimizing the number of false detections.

This problem can be formulated as a hypothesis testing between

$$\mathbf{H}_1 : o_t \sim P_{w_1}, \quad \text{for } 1 \leq t \leq T$$

and

$$\begin{aligned} \mathbf{H}_2 : o_t &\sim P_{w_1}, & \text{for } 1 \leq t \leq t_c - 1 \\ o_t &\sim P_{w_2}, & \text{for } t_c \leq t \leq T \end{aligned}$$

where $\mathbf{H}_1$ is the hypothesis that there is no abrupt change in the underlying feature distribution, and $\mathbf{H}_2$ is a composite hypothesis that an abrupt change occurs at time t_c . Below, we shall first examine an optimal offline solution which requires all feature observations and then present a customized algorithm that is suitable for online processing.

Offline Algorithm: Assuming that all of the feature observations $O = \{o_1 o_2 \dots o_T\}$ are available and independent of one another, the log-likelihood ratio, also a sufficient statistic for testing between the two hypotheses $\mathbf{H}_1$ and $\mathbf{H}_2$, is given by

$$\Lambda_T(t_c) = \log \left(\frac{\prod_{t=1}^{t_c-1} P_{w_1}(o_t) \times \prod_{t=t_c}^T P_{w_2}(o_t)}{\prod_{t=1}^T P_{w_1}(o_t)} \right) \quad (14)$$

$$= \sum_{t=t_c}^T \log \frac{P_{w_2}(o_t)}{P_{w_1}(o_t)} \quad (15)$$

where t_c is the unknown abrupt change time. The maximum-likelihood estimate of this change time is equal to

$$\hat{t}_c = \arg \max_{1 \leq t_c \leq T} \Lambda_T(t_c) \quad (16)$$

and the existence of an abrupt change can be determined by

$$d = \begin{cases} \mathbf{H}_1, & \text{if } \Lambda_T(\hat{t}_c) < \text{THD} \\ \mathbf{H}_2, & \text{if } \Lambda_T(\hat{t}_c) \geq \text{THD} \end{cases} \quad (17)$$

where THD is a threshold which depends on the priors of the hypotheses and the associated decision costs. Note that this algorithm can only be used for offline detection because all of the feature observations are required to make the decision.

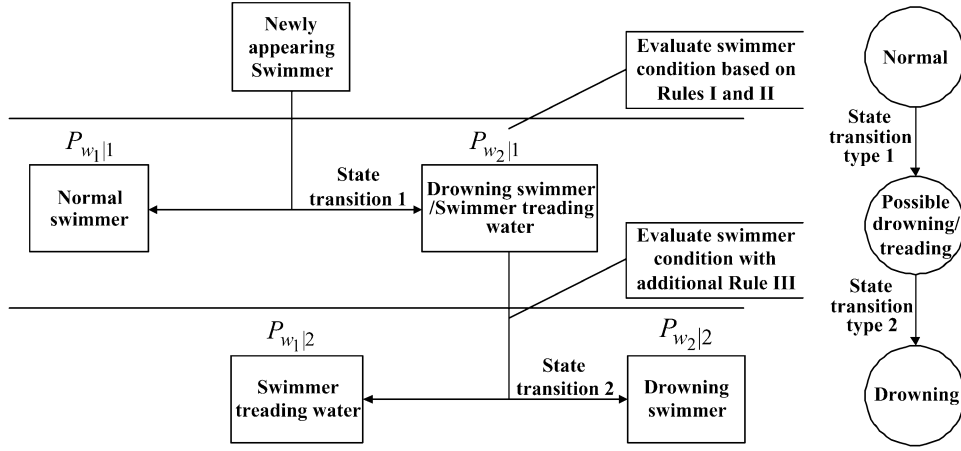
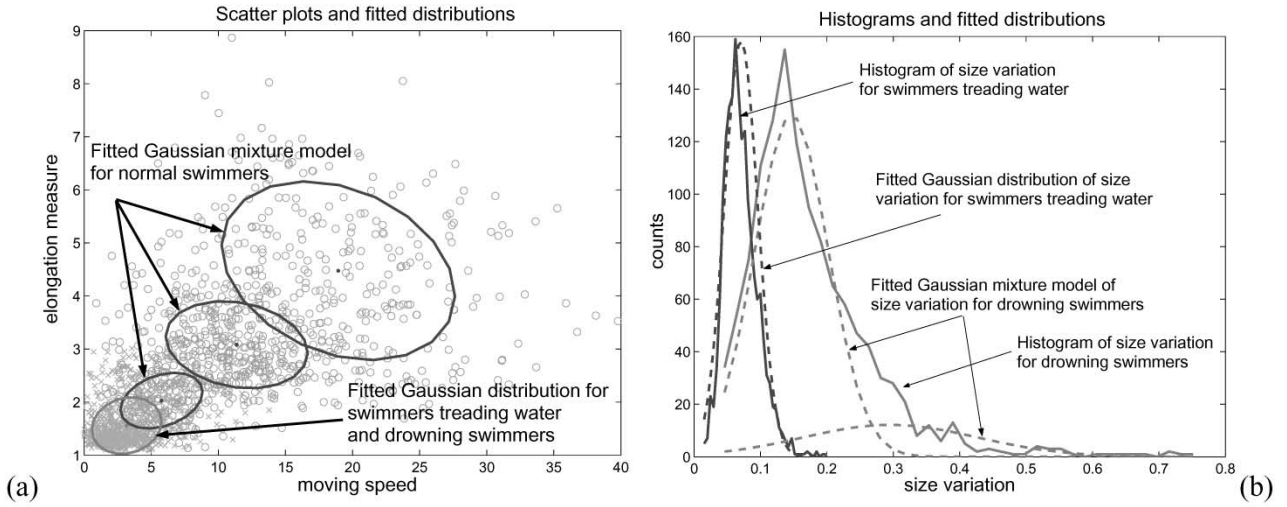


Fig. 15. Application of different rules to evaluate swimmer conditions at different states.


 Fig. 16. (a) The scatter plots and the estimated Gaussian distributions of features ($\mathcal{MS}, \mathcal{EL}$) for normal swimmers and drowning swimmers/swimmers treading water. (b) The histograms and the estimated Gaussian distributions of swimmer feature (SV) for swimmers treading water and drowning swimmers.

Online Algorithm: To develop an online algorithm for detecting the abrupt change in the underlying feature distribution as soon as possible, let us first consider the following property of relative entropy [35]:

$$E_{w_1} \left[\log \frac{P_{w_2}(o_t)}{P_{w_1}(o_t)} \right] \leq 0 \leq E_{w_2} \left[\log \frac{P_{w_2}(o_t)}{P_{w_1}(o_t)} \right] \quad (18)$$

where the left and right terms of the equation denote the expectations of the log-likelihood ratio of the feature observations under the distributions P_{w_1} and P_{w_2} , respectively. This inequality indicates that the sum of the log-likelihood ratio has a negative drift when the underlying distribution is P_{w_1} , and it has a positive drift when the underlying distribution is P_{w_2} .

Let us illustrate this property using the clip of simulated drowning considered in Section V-A. In the clip, when the swimmer lapses into a drowning situation around frame 140, his condition changes from “Normal” to “Drowning.” As a result, the underlying distribution of his moving speed and elongation measure exhibits an abrupt change, which corresponds to the type 1 state transition. Fig. 17(a) shows the log-likelihood ratio of the swimmer features (moving speed and elongation measure), while Fig. 17(b) plots the sum of the log-likelihood

ratio. We can see that the sum has a negative drift before the state transition and a positive drift afterward. Note that the maximum-likelihood estimate of the abrupt change time in this example is the turning point where the sum of the log-likelihood ratio changes from the negative drift to the positive drift.

Since only the sum of the log-likelihood ratio after the turning point is relevant to the detection of the abrupt change time, we have adopted a cumulative sum (CUSUM) procedure introduced by Page [36] as the online detection algorithm. The modified CUSUM procedure computes the sum of the log-likelihood ratio as

$$\Lambda_t = \begin{cases} 0, & \text{if } t = 0 \\ \max\{\Lambda_{t-1} + \log \frac{P_{w_2}(o_t)}{P_{w_1}(o_t)}, 0\}, & \text{if } t > 0 \end{cases} \quad (19)$$

and the existence of the abrupt change is declared at

$$\hat{t}_c = \arg \min_t \{\Lambda_t > \text{THD}\} \quad (20)$$

where Λ_t is a quantity which only starts accumulating the log-likelihood ratio after it has become positive. Fig. 18 plots the modified CUSUM of the log-likelihood ratio shown in Fig. 17(a). It should be noted that the summation profiles of

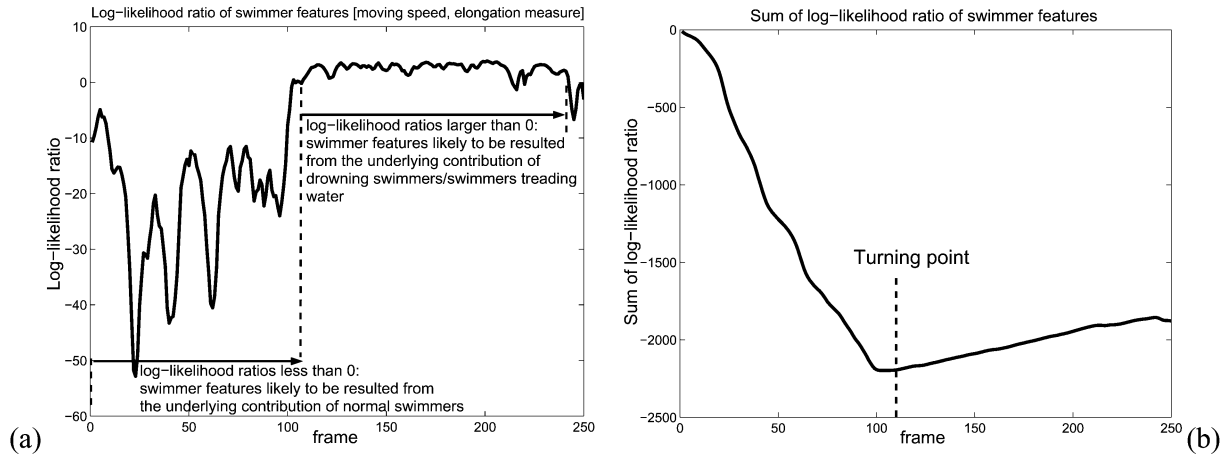


Fig. 17. (a) Plot of the log-likelihood ratio of swimmer features ($\mathcal{MS}, \mathcal{EL}$). (b) The corresponding sum of the log-likelihood ratio.

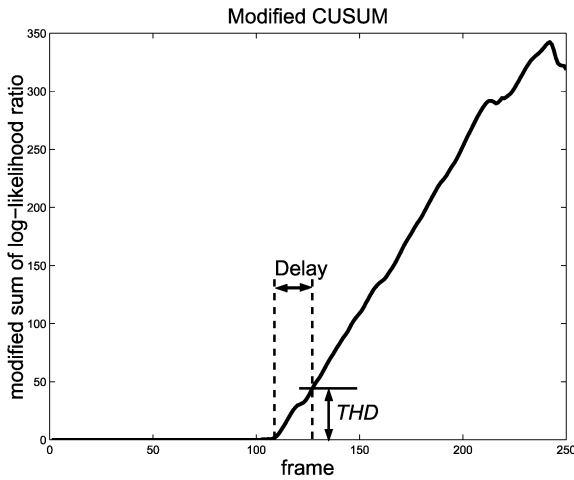


Fig. 18. Modified CUSUM of the log-likelihood ratio of the swimmer features.

the log-likelihood ratio in Figs. 17(b) and 18 after the turning point are basically the same.

E. Detection of Multiple State Transitions

There are three primary states in the state diagram shown in Fig. 14, namely “Normal,” “Possible drowning/treading,” and “Drowning” states. The transition from one state to another depends on the current state as well as the behavioral signs exhibited by the tracked swimmer. Table II lists the five possible state transitions. Generally, while a swimmer is being tracked, his/her behavioral signs and state can change from time to time. For example, the swimmer may change his/her swimming style, may tread water for some time before resuming normal swimming, and may play various water antics of which some may resemble drowning behavioral signs. Therefore, the event-inference module needs to continuously evaluate the swimmer condition and detect multiple state transitions using suitable rules and swimmer features, as summarized in Table II.

As shown in Fig. 15, the detection of a possible drowning involves two types of state transitions: the type 1 state transition from “Normal” to “Possible drowning/treading” when both Rules I and II are satisfied, and the type 2 state transition from “Possible drowning/treading” to “Drowning” when Rule III is

also satisfied. Hence, two modified CUSUM procedures are used to identify these two types of state transitions.

Detection of a Type 1 State Transition Using Rules I and II: In this case, $P_{w_1|1}$ and $P_{w_2|1}$ correspond to the distributions of $o_{t|1} = (\mathcal{MS}, \mathcal{EL})^T$ for normal swimmers ($w_1|1$) and drowning swimmers or swimmers treading water ($w_2|1$), respectively. At time t_0 , if the current state is “Normal,” a state transition to the “Possible drowning/treading” can be declared at time

$$\hat{t}_{c|1} = \arg \min_t \{\Lambda_{t|1} > \text{THD}_{I\&II}\} \quad (21)$$

where $\text{THD}_{I\&II}$ is a positive threshold and

$$\Lambda_{t|1} = \begin{cases} 0, & \text{if } t = t_0 \\ \max\{\Lambda_{t-1|1} + \log \frac{P_{w_2|1}(o_{t|1})}{P_{w_1|1}(o_{t|1})}, 0\}, & \text{if } t > t_0. \end{cases} \quad (22)$$

On the other hand, if the current state is “Possible drowning/treading,” a state transition back to “Normal” can be declared at time

$$\hat{t}'_{c|1} = \arg \min_t \{\Lambda'_t < \text{THD}_{\overline{I\&II}}\} \quad (23)$$

where $\text{THD}_{\overline{I\&II}}$ is a negative threshold and

$$\Lambda'_{t|1} = \begin{cases} 0, & \text{if } t = t_0 \\ \min\{\Lambda'_{t-1|1} + \log \frac{P_{w_1|1}(o_{t|1})}{P_{w_2|1}(o_{t|1})}, 0\}, & \text{if } t > t_0. \end{cases} \quad (24)$$

Detection of Type 2 State Transition Using Rule III: In this case, $P_{w_1|2}$ and $P_{w_2|2}$ correspond to the distributions of $o_{t|2} = \mathcal{SV}$ for swimmers treading water ($w_1|2$) and drowning swimmers ($w_2|2$), respectively. At time t_0 , if the current state is “Possible drowning/treading,” a state transition to “Drowning” can be declared at time

$$\hat{t}_{c|2} = \arg \min_t \{\Lambda_{t|2} > \text{THD}_{III}\} \quad (25)$$

where THD_{III} is a positive threshold and

$$\Lambda_{t|2} = \begin{cases} 0, & \text{if } t = t_0 \\ \max\{\Lambda_{t-1|2} + \log \frac{P_{w_2|2}(o_{t|2})}{P_{w_1|2}(o_{t|2})}, 0\}, & \text{if } t > t_0. \end{cases} \quad (26)$$

TABLE II
 FIVE POSSIBLE STATE TRANSITIONS

Current State	Next State	Criterion for transition
Normal	Possible drowning / treading	Satisfying Rules I and II
Possible drowning / treading	Normal	Not satisfying Rules I and II
	Drowning	Satisfying Rules I, II and III
Drowning	Possible drowning / treading	Not satisfying Rule III
	Normal	Not satisfying Rules I and II

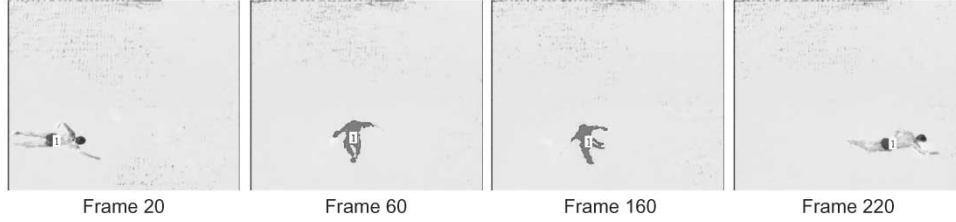


Fig. 19. Sample frames from a test clip of simulated drowning.

On the other hand, if the current state is “Drowning,” a state transition to “Possible drowning/treading” can be declared at time

$$\hat{t}'_{c|2} = \arg \min_t \left\{ \Lambda'_{t|2} < \text{THD}_{\text{III}} \right\} \quad (27)$$

where THD_{III} is a negative threshold and

$$\Lambda'_{t|2} = \begin{cases} 0, & \text{if } t = t_0 \\ \min \left\{ \Lambda'_{t-1|2} + \log \frac{P_{w_2|2}(o_{t|2})}{P_{w_1|2}(o_{t|2})}, 0 \right\}, & \text{if } t > t_0. \end{cases} \quad (28)$$

As missing a real drowning incident is more severe than triggering a false alarm (while we still want to minimize the number of false detections), and timely detection of drowning incidents is highly desired, the thresholds $\text{THD}_{\text{I\&II}}$, $\text{THD}_{\text{I\&II}}$, THD_{III} , and THD_{III} , which determine the amount of evidence to be accumulated before making a state transition, are set in such a way that entering the “Possible drowning/treading” or “Drowning” state is easier than reverting back to “Normal” or “Possible drowning/treading” state. That is, the thresholds satisfy $|\text{THD}_{\text{I\&II}}| > \text{THD}_{\text{I\&II}}$ and $|\text{THD}_{\text{III}}| > \text{THD}_{\text{III}}$. Although this setting may lead to some false alarms, it is in line with our objective of assisting the lifeguards on duty by drawing their attention to possible drowning swimmers as soon as the evidence suggests so. (In our implementation, $\text{THD}_{\text{I\&II}}$ and $\text{THD}_{\text{I\&II}}$ are set to 40 and -60 , while THD_{III} and THD_{III} are set to 1 and -3 , respectively.)

Let us illustrate the detection of multiple state transitions using a test video of simulated drowning. Sample frames of this test video are shown in Fig. 19. In the video, the swimmer starts simulating drowning around frame 50 and struggles on the water surface till about frame 200 when he resumes normal swimming. Fig. 20(a) depicts how the modified CUSUM procedure detects the change in the swimmer condition using Rules I and II, i.e., based on (21) and (22). When the swimmer stops moving continuously and his/her body becomes nearly vertical in the

water, the CUSUM of the log-likelihood ratio of the swimmer’s moving speed and elongation measure exceeds the threshold $\text{THD}_{\text{I\&II}}$ at frame 45. Hence, the state of the swimmer changes from “Normal” to “Possible drowning/treading.”

From frame 45 onward, the module restarts accumulating the log-likelihood ratio of swimmer observations based on (23) and (24) to test whether the swimmer remains in the “Possible drowning/treading” state. Meanwhile, the module also evaluates the swimmer’s condition using Rule III, i.e., based on (25) and (26) as shown in Fig. 20(b). As the struggle of the swimmer causes large size variations within a short duration, the CUSUM of the log-likelihood ratio of the swimmer’s size variation soon exceeds the threshold THD_{III} , which puts the swimmer into the “Drowning” state. Once the swimmer resumes normal swimming, the module also successfully detects that the swimmer’s is no longer in danger, i.e., the state transits back to “Normal.” This example should shed some light on the effectiveness of the modified CUSUM procedure in continuously monitoring the swimmer condition and making timely decisions.

VI. EXPERIMENTAL RESULTS

We present in this section some representative results of the experiments that we have conducted for detecting possible drowning incidents and monitoring swimmers in public pools. The results are discussed in two subsections, which are devoted to the vision component and the event-inference module, respectively.

A. Experimental Results of the Vision Component

We tested our vision component in two types of swimming pools, competition pools and teaching pools. The overhead camera used to monitor swimmers was high-mounted on one side of the pools in order to cover a large field of view and to minimize the occlusions among swimmers. To test the robustness of the vision component, the video data were recorded

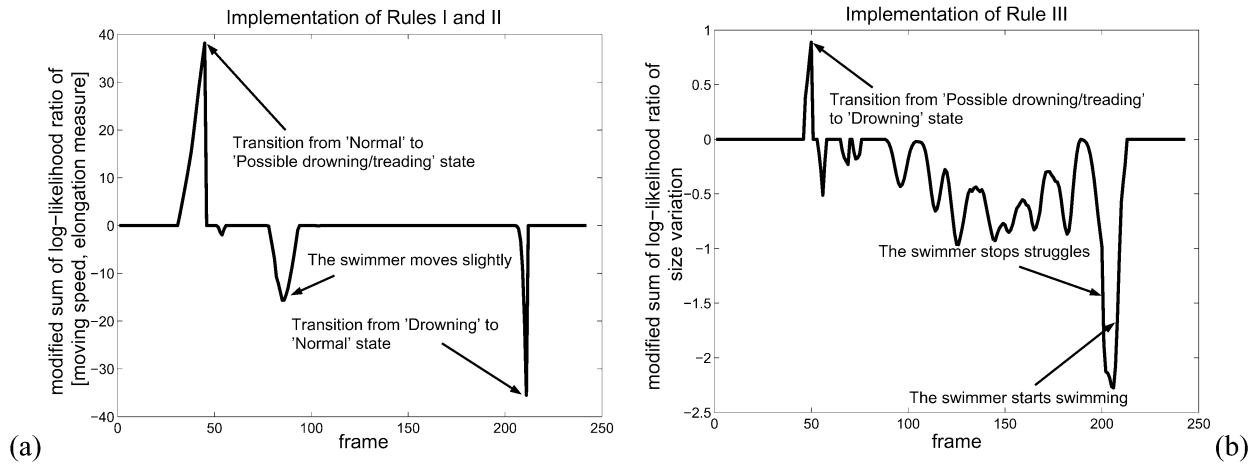


Fig. 20. Implementation of the modified CUSUM procedure using: (a) Rules I and II and (b) Rule III.

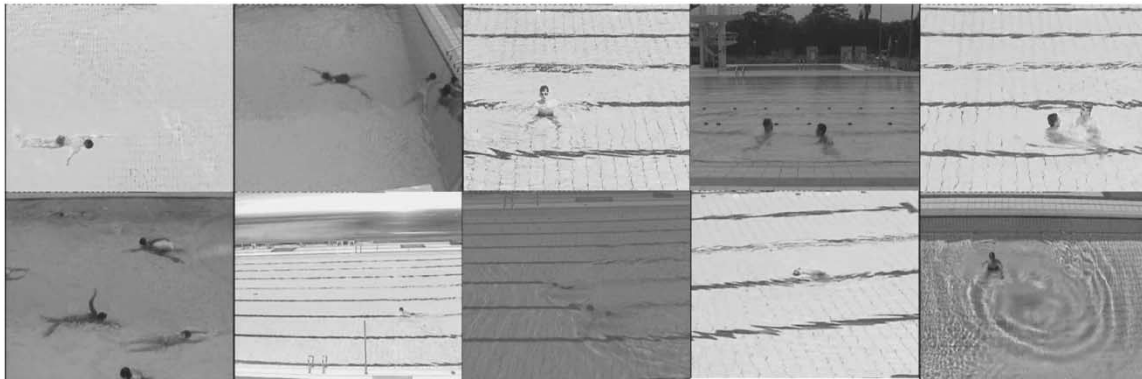


Fig. 21. Sample frames of video clips used to test the vision component.

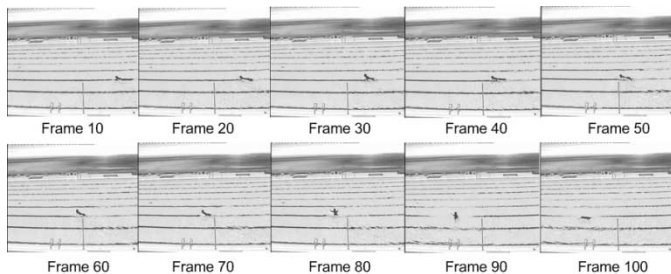


Fig. 22. Sample frames of a challenging test video clip.

with different camera zooming lengths and viewing angles, and about 1 h in total of video data were captured under different weather and lighting conditions. In many of the test video clips, the swimmers were not given any instruction on swimming styles or moving directions, etc. Fig. 21 shows some sample frames of the video data used to test the vision component.

As both types of swimming pools satisfy the assumption made in the proposed vision component, i.e., the background scene of a swimming pool is rather homogeneous, consisting of only a few distinct scene classes, the vision component can reliably detect and track swimmers within the monitored pool areas. Fig. 22 shows a challenging scenario, in which the black strip on the pool floor overlaps with the moving trajectory of a swimmer. The swimmer's movement caused drastic change in

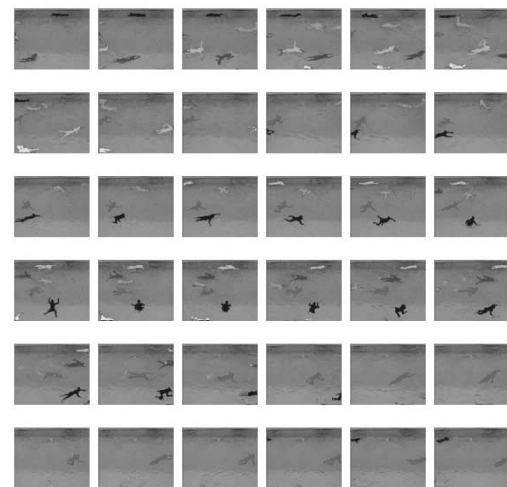


Fig. 23. Sample frames of a test video clip with multiple swimmers.

the background scene and the color of his swimming trunk was similar to that of the black strip. The vision component still managed to well segment the swimmer, indicated in dark color, without including motion clutters.

Fig. 23 shows the sample frames of another test video, in which the swimmers are highlighted in different intensities by the vision component to differentiate their identities. In the

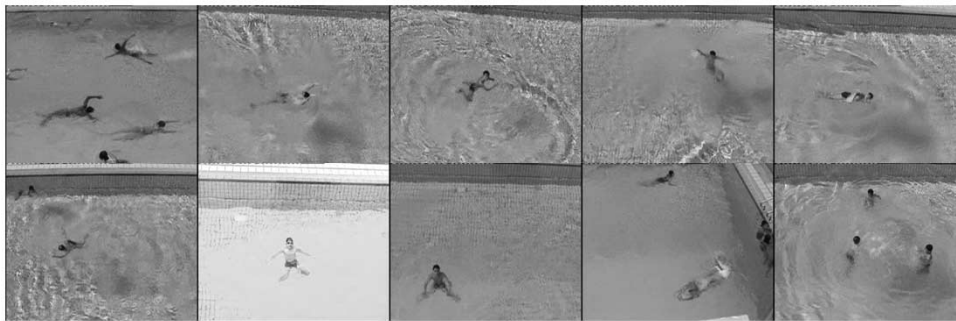


Fig. 24. Sample frames of video clips used to test the event-inference module.

TABLE III
SUMMARY OF CLIPS IN THE TRAINING SET

Training Clip Category	Drowning	Normal				Treading water
	simulated drowning	back stroke	breast stroke	crawl stroke	butterfly style	treading water
Subject 1	3	1	1	1	1	2
Subject 2	2	1	1	1	1	2
Subject 3	3	0	1	1	1	2

video, the swimmers entered the monitored pool area from different regions, swam in different styles and directions, and had some short-duration collisions due to rather high swimmer population. As the vision component did not attempt to infer the identities of the swimmers involved in a collision, the swimmer identities, indicated by different intensities could sometimes change after collision.

It should be noted that the parameter settings of the proposed vision component remained unchanged while processing all the test videos. In most cases, the vision component was able to reliably detect and track all the swimmers and extract their motion and shape features. However, when the swimmers were very close to each other in some occasions, the vision component blindly tracked the swimmers as a merged moving region until they separated. In such case, the extracted features carried little information about the conditions of these swimmers. Currently, the event-inference module does not evaluate the conditions of such occluded swimmers, as a drowning incident can likely be recognized by other swimmers when they come close.

B. Experimental Results of the Event-Inference Module

When developing the event-inference module, we faced sheer difficulty in collecting video data of drowning incidents. It is neither ethical nor feasible to obtain enough real drowning clips for training and testing purposes. Hence, in our experiments, we invited three volunteers, who were healthy graduate students with some swimming experience, to simulate the motions of drowning swimmers recorded in a documentary video “On Drowning” [37], which has been used by the American Red Cross for lifeguard training, under the guidance of a professional lifeguard. In total, we captured more than 30 min of video data of simulated drowning, which were then digitized with a resolution of 352×288 pixels. Although the digitized video frame rate was 30 frames/s, only one out of every four consecutive frames

was processed in order to reduce the computational cost. Therefore, the effective processing rate of the proposed approach is about 7.5 frames/s.

The digitized video data were manually segmented into 16 clips with lengths ranging from 700 to 1200 frames. Each clip contained only one drowning incident. Besides video clips of drowning simulation, we also captured another 45-min video of the volunteers swimming in different styles and playing various water antics, including treading water and splashing water. Sample frames of our video data are shown in Fig. 24 to illustrate the variety of the video data.

The video data were grouped into a training set and a test set. The training set (listed in Table III) was used to learn and model the feature distributions under different swimmer conditions, as described in Section V-C, while the test set (listed in Table IV) was used to assess the performance of the proposed system in detecting possible drowning incidents.

The overall performance is summarized in Table V. Out of the eight simulated drowning, 44 normal swimming, and 17 treading water events, the proposed approach was able to detect all of the simulated drowning incidents and did not make any wrong judgment for normal swimming events. However, there are two events of swimmer treading water misjudged as possible drowning incidents, which are listed in Table VI.

Fig. 25 shows the sample frames of these two misclassified treading water events. In clip “Std_tread3,” the swimmer played water antics by kicking the water intensively. The swimmer motion actually satisfied the three rules formulated for identifying possible drowning swimmers. In clip “Std_tread4,” the swimmer with label 1 rotated his body rapidly within a small neighborhood, which can be seen in comparison to the swimmer with label 2 moving in a crawl stroke. This led to a large size variation within a short duration. Exploring other features or methods for handling such failure cases is one possible future direction.

TABLE IV
DETAILS OF CLIPS IN THE INDEPENDENT TEST SET

Clip	Length of Event (seconds)	Number of Events
Std_drown1	22	1 drowning incident simulated by Subject1
Std_drown2	25	1 drowning incident simulated by Subject1
Std_drown3	30	1 drowning incident simulated by Subject1
Std_drown4	15	1 drowning incident simulated by Subject2
Std_drown5	16	1 drowning incident simulated by Subject2
Std_drown6	18	1 drowning incident simulated by Subject2
Std_drown7	12	1 drowning incident simulated by Subject3
Std_drown8	16	1 drowning incident simulated by Subject3
Multiple1	108	18 normal swimmers in different styles and 3 swimmers treading-water
Multiple2	122	12 normal swimmers in different styles and 2 swimmers treading-water
Multiple3	127	5 normal swimmers in different styles and 1 swimmers treading-water
Multiple4	136	6 normal swimmers in different styles and 2 swimmers treading-water
Multiple5	270	3 normal swimmers in different styles and 3 swimmers treading-water
Std_tread1	71	1 treading-water (Subject1)
Std_tread2	32	1 treading-water (Subject1)
Std_tread3	67	1 treading-water (Subject2)
Std_tread4	45	1 treading-water (Subject2)
Std_tread5	70	1 treading-water (Subject3)
Std_tread6	50	1 treading-water (Subject3)

TABLE V
OVERALL DROWNING DETECTION PERFORMANCE

Event Category	Training set			Test set		
	No. of events	False	Missed	No. of events	False	Missed
Drowning	8	0	0	8	0	0
Normal Swimming	11	0	0	44	0	0
Treading Water	6	0	0	17	2	0

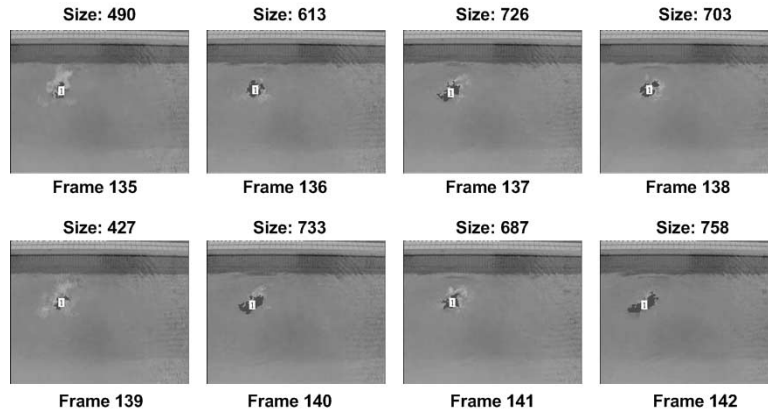


Fig. 25. Sample frames of false detection in the clip “Std_tread3.”

As another primary objective of our approach is to detect drowning incidents at the earliest possible stage, we also evaluated the delay incurred before affirming the occurrence

of each drowning incident. Table VII lists the detection delay of the proposed approach as compared to the judgment of an experienced lifeguard. It can be seen from the table that

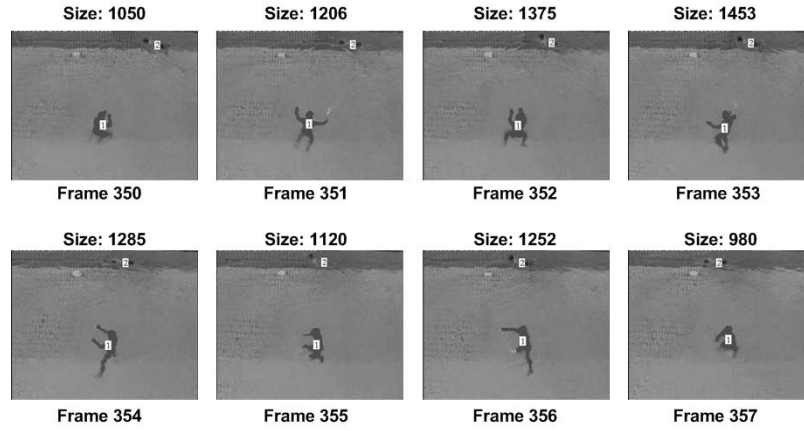


Fig. 26. Sample frames of false detection in the clip “Std_tread4.”

TABLE VII
DETECTION DELAY INCURRED BEFORE THE OCCURRENCE OF EACH DROWNING INCIDENT IS AFFIRMED

Clip	Actual Drowning Frames	Detected Drowning Frames	Detection Delay (seconds)
Std_drown1	56-220	71-232	2
Std_drown2	49-240	64-251	2
Std_drown3	315-530	318-546	0.4
Std_drown4	24-120	38-129	2
Std_drown5	12-70	27-84	2
Std_drown6	10-105	21-109	1.5
Std_drown7	10-95	30-105	2.7
Std_drown8	85-138	90-145	0.7

TABLE VI
TWO FALSE DETECTIONS

Clip	Treading Duration	False Detection
Std_tread3	100-201	131-153
Std_tread4	120-454	330-403

the modified CUSUM procedure can quickly detect all of the drowning incidents. The minimum delay is only three processing frames (less than 1 s), while the maximum delay is 20 processing frames (about 3 s). The latter incurred a longer delay because the swimmer, who simulated the drowning, moved in a constant direction while struggling; the swimmer is considered lapsed into drowning only when the swimmer was eventually confined within a small neighborhood.

VII. CONCLUDING REMARKS

In this paper we have presented a novel vision-based approach to early detection of drowning incidents in swimming pools. Different from other existing systems, the proposed approach makes use of only overhead cameras to monitor swimmers and aims at detecting drowning incidents at the earliest possible stage.

The approach consists of two main parts: a vision component using an overhead camera to detect and track swimmers and an

event-inference module using a sequential change detection algorithm to parse the sequences of swimmer features for early drowning behavioral signs. The event-inference module is constructed based on three reasoning rules formulated from the universal motion characteristics of drowning swimmers. The rules are systematically integrated with a finite state machine, where each state transition corresponds to a change of swimmer condition or a confirmation of a possible drowning incident.

One major difficulty faced in developing our drowning detection approach is the lack of video data of real drowning incidents. Hence, we have chosen to rely on video data of simulated drowning to perform the training and testing. The proposed approach has been tested with a variety of independent test videos of simulated drowning, and very encouraging results, as compared to the judgments made by an experienced lifeguard, have been obtained.

As for future work, one direction is to identify a richer set of features which can more effectively differentiate drowning swimmers from swimmers treading water, hence reducing the number of false alarms. However, such reduction of false alarms should be carefully applied so that it does not affect the detection performance, as missed detection of a real drowning incident may result in an irremediable loss. Also, in order to have complete coverage of a large swimming pool, the integrated use of multiple cameras is to be examined. Furthermore, although each swimmer is an autonomous moving object, his/her motion characteristics are in fact subject to the influence from the sur-

rounding environment. For example, when the swimmer population in the pool is high, it is more likely to observe swimmers treading water. It is also important to study the effects of other external causes (such as water temperature, light, and time) of drowning incidents. All of these factors, which have possible causal relations to drowning incidents, should also be considered together with the swimmer motion and shape features and could probably be incorporated into a Bayesian network [38].

In summary, our approach makes a novel attempt to evaluate swimmers' conditions by analyzing their motion and shape features and provides solutions to several key problems in detecting possible drowning incidents. While challenging in many aspects, a successful system could help save lives.

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